KUBEFLOW

Machine Learning Operations, Reimagined

An open-source platform for building, training, deploying, and managing ML workflows on Kubernetes.

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What is Kubeflow?

Kubeflow is an open-source platform designed to make Machine Learning workflows easier to develop, deploy, and manage on Kubernetes.

BUILT ON KUBERNETES

Uses Kubernetes as the underlying infrastructure.

PORTABLE WORKFLOWS

Supports cloud, on-premises, and hybrid environments.

THE MAIN GOAL

Simplify development, orchestration, training, deployment, and management across the ML lifecycle.

SCALABLE REPRODUCIBLE ML LIFECYCLE

Why Kubeflow is Needed

Traditional ML workflows can become difficult to maintain as projects grow.

01 Manual training

Training and deployment require repeated human effort.

02 Complex workflows

Many connected tasks are difficult to coordinate.

03 Environment mismatch

Development and production may differ.

04 Scaling limits

Large workloads need more compute resources.

05 Reproducibility

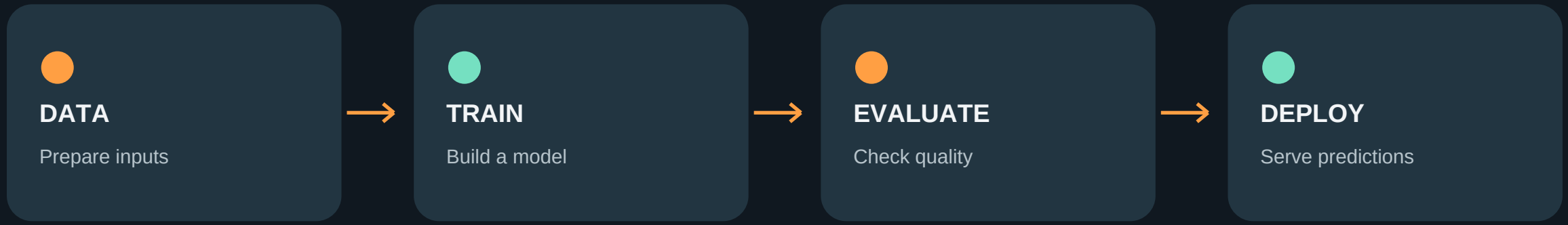
Repeated experiments may be hard to recreate.

06 Dependencies

Managing libraries and resources adds complexity.

The Kubeflow Solution

Kubeflow provides a Kubernetes-based platform to automate and manage ML tasks in a scalable and reproducible manner.



AUTOMATION SCALABILITY REPEATABILITY

Kubeflow and Kubernetes

KUBERNETES

An open-source container orchestration platform. It automates deployment, scaling, and management of containerized applications.

KUBEFLOW

Adds ML-specific tools and workflows that run on Kubernetes.

Key benefits

- **Scalability**
- **Containerization**
- **Resource management**
- **Portability**
- **Automated deployment**

The ML workflow at a glance

Kubeflow connects the key stages of an ML project into one manageable workflow.

01 DEVELOP	**Kubeflow Notebooks**	Explore data and write code
02 ORCHESTRATE	**Kubeflow Pipelines**	Connect reusable workflow steps
03 TRAIN & TUNE	**Training + Katib**	Train models and tune parameters
04 SERVE	**KServe**	Expose models for inference

Major Kubeflow Components

Pipelines

Build and automate ML workflows.

Notebooks

Interactive development environments.

Training

Scalable and distributed training jobs.

Katib

Automated hyperparameter tuning.

KServe

Model serving and inference.

Model Registry

Organize and track ML models.

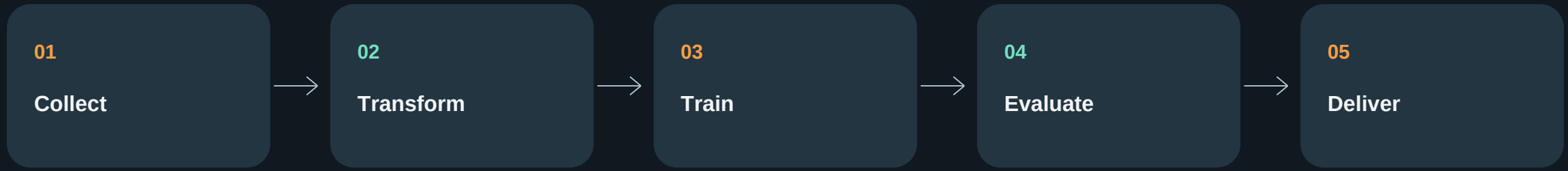
Dashboard

UI for interacting with Kubeflow resources.

Kubeflow Pipelines

Kubeflow Pipelines is a platform for building and running machine-learning workflows.

An ML workflow is divided into reusable components. Each component performs a specific task, and the components connect to form an automated pipeline.



Pipelines can be re-run with different parameters to support repeatable experiments.

From Data to Deployment

A typical ML lifecycle can be represented as a sequence of repeatable stages.

WHY IT HELPS

Workflow automation, reproducibility, reusable components, and easier experiment management.

Kubeflow Notebooks

WHAT THEY ARE

Interactive development environments within Kubernetes. Data scientists can create and manage notebook environments.

TOOLS & RESOURCES

Supports tools such as JupyterLab and lets users select compute resources for their workloads.

Common uses

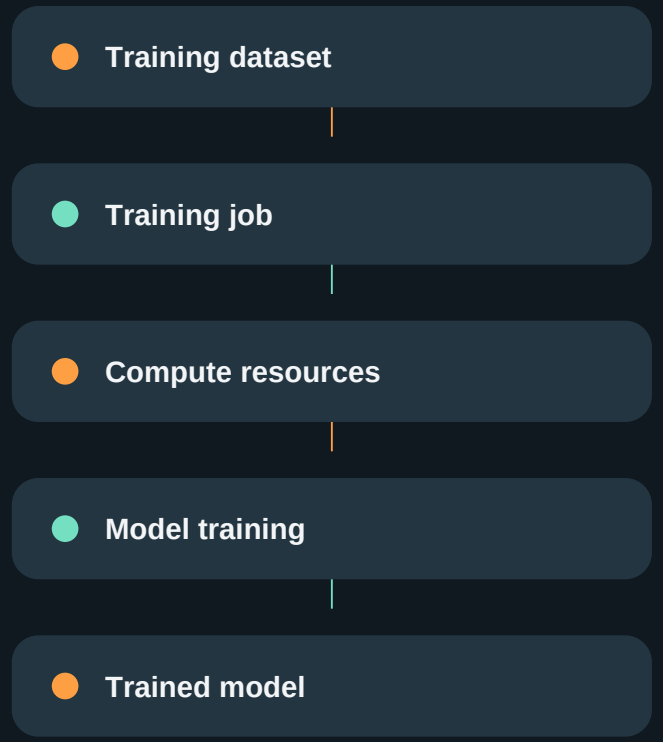
- Data exploration
- Data preprocessing
- Model development
- Experimentation
- Testing ML algorithms

Benefit: container isolation provides a consistent and manageable ML development environment.

Kubeflow Training

Kubeflow Training provides tools for running ML training workloads on Kubernetes.

- Supports single-node and distributed training.
- Jobs can use CPU, GPU, and other compute resources.
- Kubernetes manages the resources required for training.



Why scalable training matters

Kubeflow helps teams use computing infrastructure efficiently as training needs grow.

01	**Scalable training**	Increase resources as workloads grow.
02	**Distributed training**	Train across multiple workers when needed.
03	**Resource management**	Coordinate compute resources through Kubernetes.
04	**Infrastructure efficiency**	Use available computing capacity more effectively.

Kubeflow in MLOps

Kubeflow supports important stages of the Machine Learning lifecycle.

Development	→	**Kubeflow Notebooks**
Workflow automation	→	**Kubeflow Pipelines**
Model training	→	**Kubeflow Training**
Experimentation	→	**Katib**
Model deployment	→	**KServe**
Infrastructure	→	**Kubernetes**

MAIN BENEFIT Moves ML models from development to production in an automated, scalable way.

Advantages & Applications

ADVANTAGES

- Open-source and flexible
- Scalable ML workloads
- Automates workflows
- Reproducible experiments
- Works with Kubernetes
- Supports training and deployment

APPLICATIONS

- Computer Vision
- Natural Language Processing
- Recommendation Systems
- Predictive Analytics
- Fraud Detection
- Large-scale Machine Learning

Conclusion

- Kubeflow is an open-source platform for Machine Learning on Kubernetes.
- It provides tools for development, training, workflow automation, and deployment.
- It reduces manual work in ML workflows.
- It enables scalable and reproducible ML processes.
- It connects ML development with MLOps and production environments.

“Kubeflow makes Machine Learning workflows easier to automate, scale, and manage using Kubernetes.”

Thank You

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Questions & Discussion